MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 12 (8-12 Nov 2026) - Period 1	Unit 4 - 4.1 Setting the Game Elements (2/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 12 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 70% ordered listener correctly; sessionStorage understood by 80%. Today: loops and asynchronous timing for the game.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use for/while loops and asynchronous methods (setTimeout, setInterval, clearInterval) to repeat actions and time events. Sprint 4.1 obj. 3-4; Challenges 2-3; CSTA 3A-AP-15.

Success criteria — *what will successful learning look / sound like?*

I write a for loop to repeat code; I use setTimeout/setInterval with milliseconds correctly; I stop a timer with clearInterval; my countdown works.

Key knowledge / skill / understanding:

Loops repeat; 1 s = 1000 ms; asynchronous code runs later, not in order.

Likely misconception / access or challenge need:

setTimeout pauses the program; loops always need a counter from 0.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*	ACTIVITY 2 *student task + evidence of learning*	ACTIVITY 3 *student task + evidence of learning*
Strategy: Inquiry / investigation **Time:** 8 min	**Strategy:** Problem solving **Time:** 20 min	**Strategy:** Cooperative learning **Time:** 8 min
What students will do: Predict-then-run: 4 short code snippets (loop, setTimeout order). Students predict console output, run, then explain differences.	**What students will do:** Challenge 2: countdown timer from 10 that disappears at 0 and shows a birthday message (or Eid/National Day greeting); then add a loop creating 9 holes dynamically.	**What students will do:** Pairs swap code and trace each other's timer with a trace table (time, value, output).
Evidence tool: Mini-whiteboards Assessment: ☑ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group	**Evidence tool:** Practical / performance task Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☑ Individual ☐ Pair ☐ Group	**Evidence tool:** Peer-assessment checklist Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group
Resources: Snippets in browser console.	**Resources:** VS Code; QR 'Asynchronous methods'; source files.	**Resources:** Trace table template.
Support (low performers / SEN): Annotated snippets; buddy support.	**Support (low performers / SEN):** Countdown starter with comments; flowchart.	**Support (low performers / SEN):** Partially filled trace table.
Stretch (fast learners / gifted): Explain why setTimeout(...,0) still runs last.	**Stretch (fast learners / gifted):** Challenge 3: print primes up to n using loops and if-else.	**Stretch (fast learners / gifted):** Add an edge case (n = 0 or negative) to test.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 7 min

Hook: a countdown on the board - how do games keep time? Model a for loop with a trace table; model setInterval with a counter and clearInterval; convert seconds to ms on the board. Share success criteria.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students choose the celebration message (birthday, Eid, Diwali, National Day) - culturally responsive context; trace tables build computational thinking.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Infinite loops freezing the tab; forgetting clearInterval; using seconds not ms; counter declared inside the interval. Response: 'my timer goes negative'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: write a setInterval countdown from 5 in pseudocode; convert 2.5 s to ms. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: finish countdown; try the primes challenge (optional). Review student book 4.2 'Start the game'.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Predict-then-run created real surprise and talk. Celebration choice made it personal and inclusive. 80% built a working countdown.

What will I change next time?

Show how to recover from an infinite loop (close tab) before practice. Trace tables need a worked example first.

Students needing follow-up:

Students who froze browsers repeatedly (10/4, 10/5) need loop-condition practice. Inform parents of 5 students behind on submissions.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☐3 ☑C ☐H	☑1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
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ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 12 (8-12 Nov 2026) - Period 2	Unit 4 - 4.2 Core Logic of the Game (1/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 12 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Countdowns working for ~80%; ms conversion now secure. Today: start the game, make worms pop out at random times in random holes.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students build core game logic: start the game, make worms pop out, set random times (Math.random range formula) and choose random holes. Sprint 4.2 obj. 1-4; CSTA 3A-AP-16; Activity 1.

Success criteria — *what will successful learning look / sound like?*

I start the game on a button click; I generate a random time between min and max; I choose a random hole; worms pop out repeatedly.

Key knowledge / skill / understanding:

Math.random() gives 0 to <1; min + rand*(max-min); Math.floor for whole numbers; array of holes.

Likely misconception / access or challenge need:

Math.random gives whole numbers; same worm can't appear twice in the same hole.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Problem solving **Time:** 8 min

What students will do:

Activity 1: calculate a random time between 2000 and 3000 ms for rand = 0.83 by hand, then verify in Chrome's Console.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Chrome Console; formula slide.

Support (low performers / SEN):

Calculator; worked example with 0.5.

Stretch (fast learners / gifted):

Write a function randomBetween(min,max) and test 5 values.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Code along: startGame(), randomTime(), randomHole() and popOut() using setTimeout; worms appear in random holes until the game ends.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; project source 'Core logic'; student book 4.2.

Support (low performers / SEN):

Code skeleton with functions named; pair programming.

Stretch (fast learners / gifted):

Prevent the same hole twice in a row (compare with lastHole).

ACTIVITY 3 *student task + evidence of learning*

Strategy: Debate **Time:** 7 min

What students will do:

Is a game fair if it is random? Teams argue for/against using real examples (dice, sports draws) and suggest how to make randomness feel fair.

Evidence tool: Oral questioning

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Debate prompt; examples.

Support (low performers / SEN):

Sentence starters.

Stretch (fast learners / gifted):

Link to probability: chance of the same hole twice with 9 holes.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: roll a dice - how can a computer 'roll'? Model Math.random, the range formula, Math.floor with an array of holes. Show the flow: start -> random hole -> pop out -> hide -> repeat. Share success criteria.

What is new or innovative in this lesson? *(ETEC looks for this)*

Numeracy and probability embedded in coding; students debate fairness of randomness, linking maths to game design ethics.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☐ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Using Math.round giving index 9 (out of range); setTimeout not called again; CSS class for 'up' worm missing. Likely: 'worms appear all at once'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: compute min + 0.25*(max-min) for 1000-4000; why Math.floor not Math.round? Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: finish pop-out logic; upload a 20-second screen recording or screenshot. Optional: Challenge 3 Start/Stop random colour.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Calculating by hand first made the code make sense; probability debate engaged quieter maths-strong students. 75% had worms popping by the end.

What will I change next time?

Show the off-by-one error deliberately next time. Give a skeleton to all students to save typing time.

Students needing follow-up:

Students whose worms don't hide (CSS class issue) - approx. 3 per class; peer mentors assigned for next lesson.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

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